Tutorial for Chicken Behavior Functions in MATLAB

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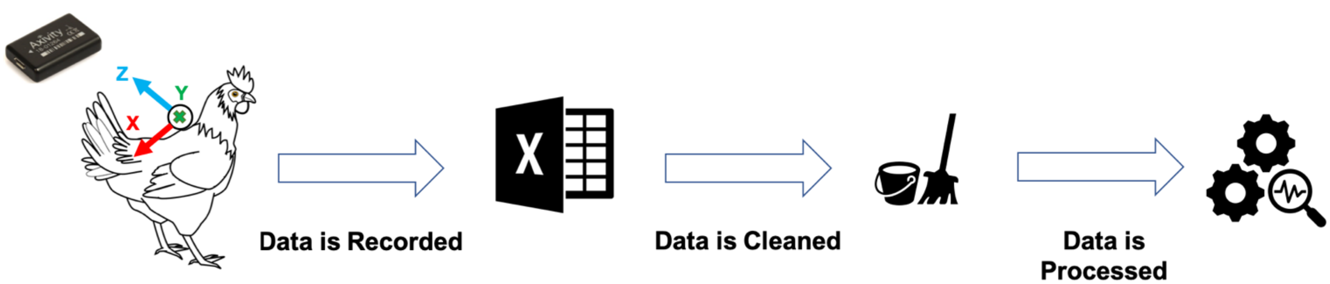
This tutorial describes the library of key functions, developed in MATLAB, to identify chicken behaviors: pecking, preening, and dustbathing. The classification algorithm was developed based on combination of shapes and features as discussed in Abdoli et al. (2020) and was used for data analyses in Murillo et al. (2020).

1) ***Pecking:*** bringing the beak to the ground, striking at the ground
2) ***Preening:*** manipulating, rearranging, pulling, or smoothing body feathers by the beak
3) ***Dustbathing:*** bird is in a sitting or lying position with feathers raised in a vertical wing-shake, including feather-ruffling and shaking (Daigle et al. 2014).

The process of collecting data from measurement sensors, preparation and processing of data is shown in following figure and discussed in (Abdoli et al. 2019). We have three main functions, with two of the functions for cleaning and preparation of data (***daySlicer*** and ***chicklet*** functions) before we can perform classification. Following preparation of data, we can proceed with classification task (***behaviorSearch*** function) which returns the number of pecking, preening and dustbathing behaviors in the input dataset.



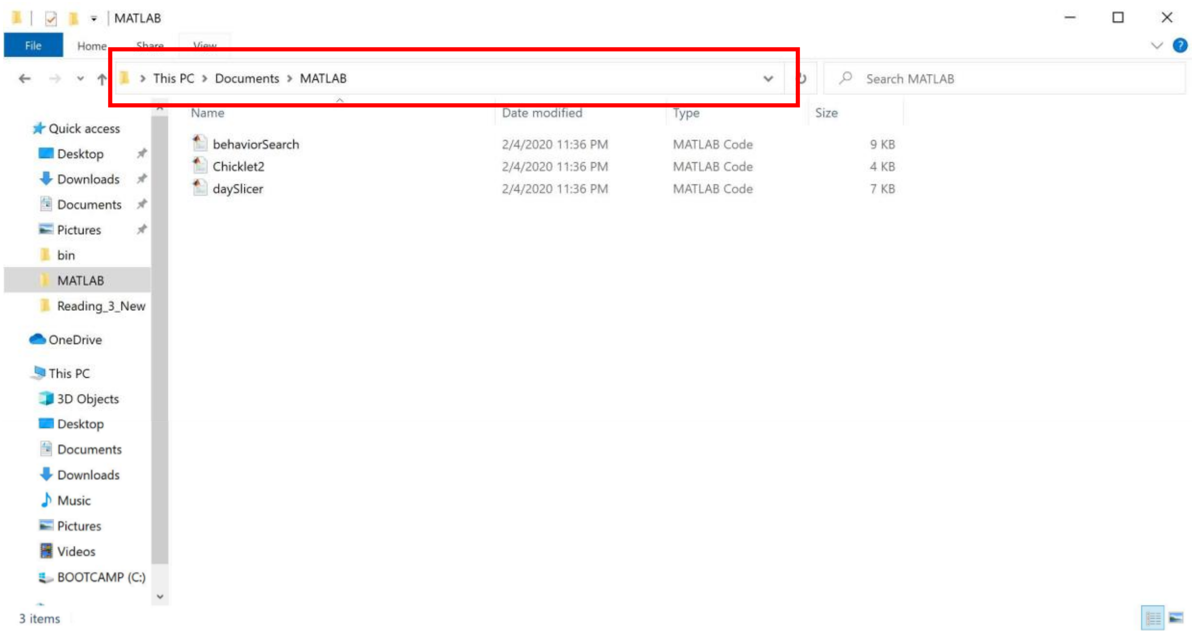
In this research we utilized Axivity AX3 sensor (website: http://axivity.com/product/ax3), recording data at default frequency of 100 Hz. The data files were formatted into four columns (***TimeStamp, X-Axis, Y-Axis, Z-Axis***). The functions should work on any platform and operating system as long as you have MATLAB software installed.

Below is a list of revisions made to the algorithm:

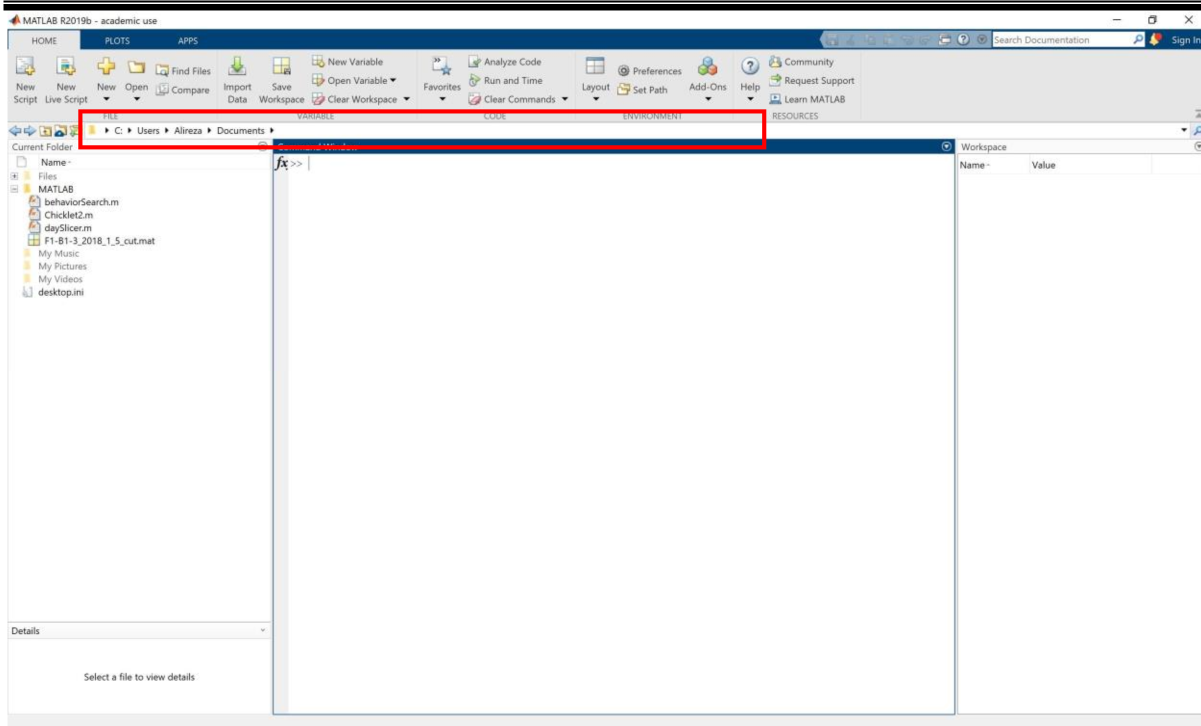
Revisions	Date	Notes
V1	2018	Shape based classification (Abdoli et al. 2018).
V2	2020	Shape / Feature based classification (Abdoli et al. 2020).

Initial Setup:

In order to get started please make sure that after you download data and function files, change your MATLAB path to the same folder you have downloaded the data and files. This way the downloaded files will be visible to MATLAB.



The next step is to open an instance of MATLAB program. By default, MATLAB looks into the Documents\MATLAB folder.



1. ***daySlicer*****:**

Large CSV files may be challenging to work with in MATLAB. It may be more convenient to work with individual days of data. ***The "daySlicer.m" functions takes in a single CSV file along with the start and end dates; then it breaks the CSV file into day-long MAT files*** (native to MATLAB) which are much smaller in size and easier to process in MATLAB. The function is invoked as follows:

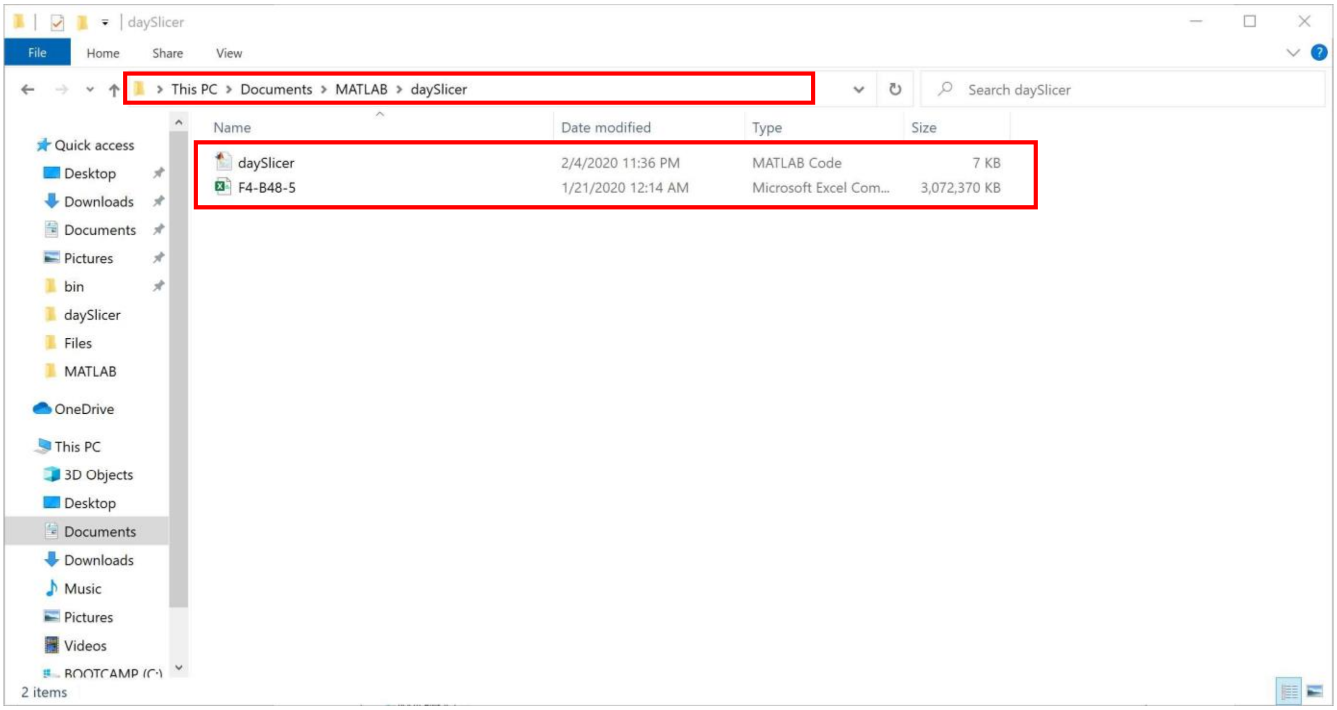
daySlicer ('2018-03-29 00:00:00', 11)

Input: *(1) Starting time-stamp (**Required**) (2) Number of days to extract from within the CSV file (**Optional**)(**Default: 1 Day**).*

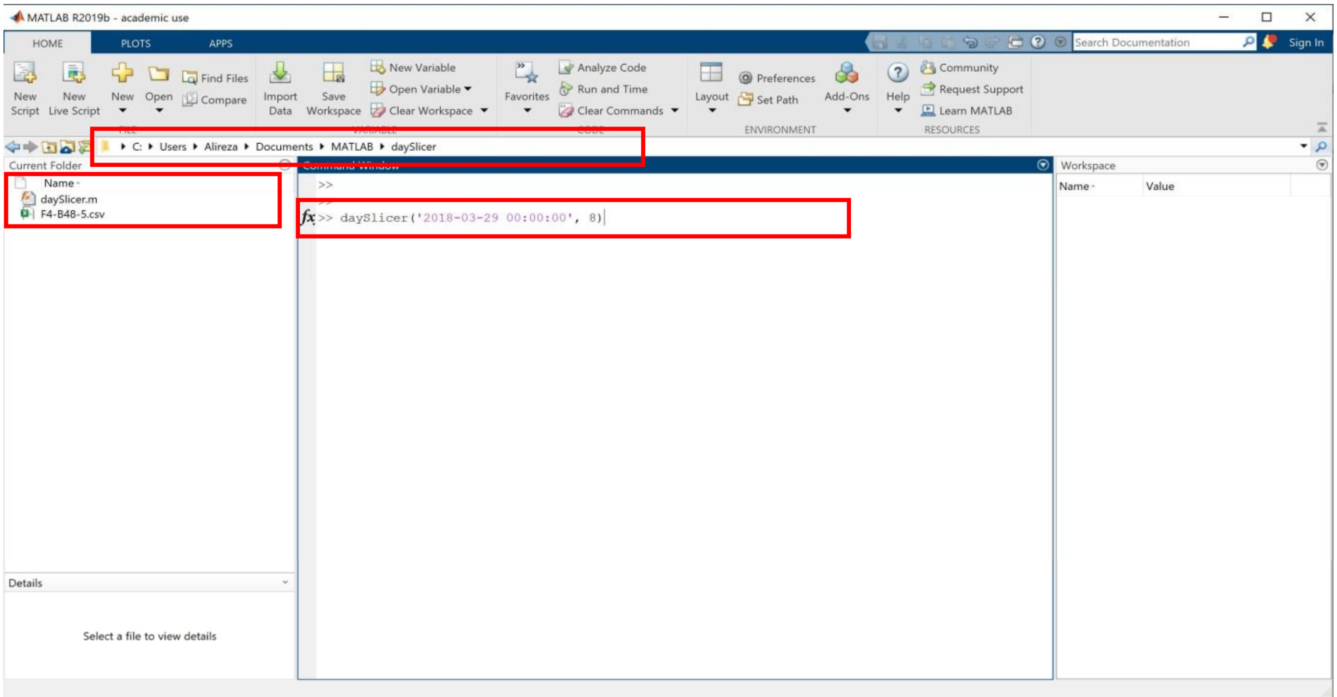
Output: *Based on input argument (2), some number of 24-hours MAT data files will be extracted and saved to the same directory as the original CSV file.*

Please put the "*daySlicer.m*" function along with any number of CSV files you have for slicing in the same folder. Change your MATLAB path to the newly created folder and run the *daySlicer* function as discussed above.

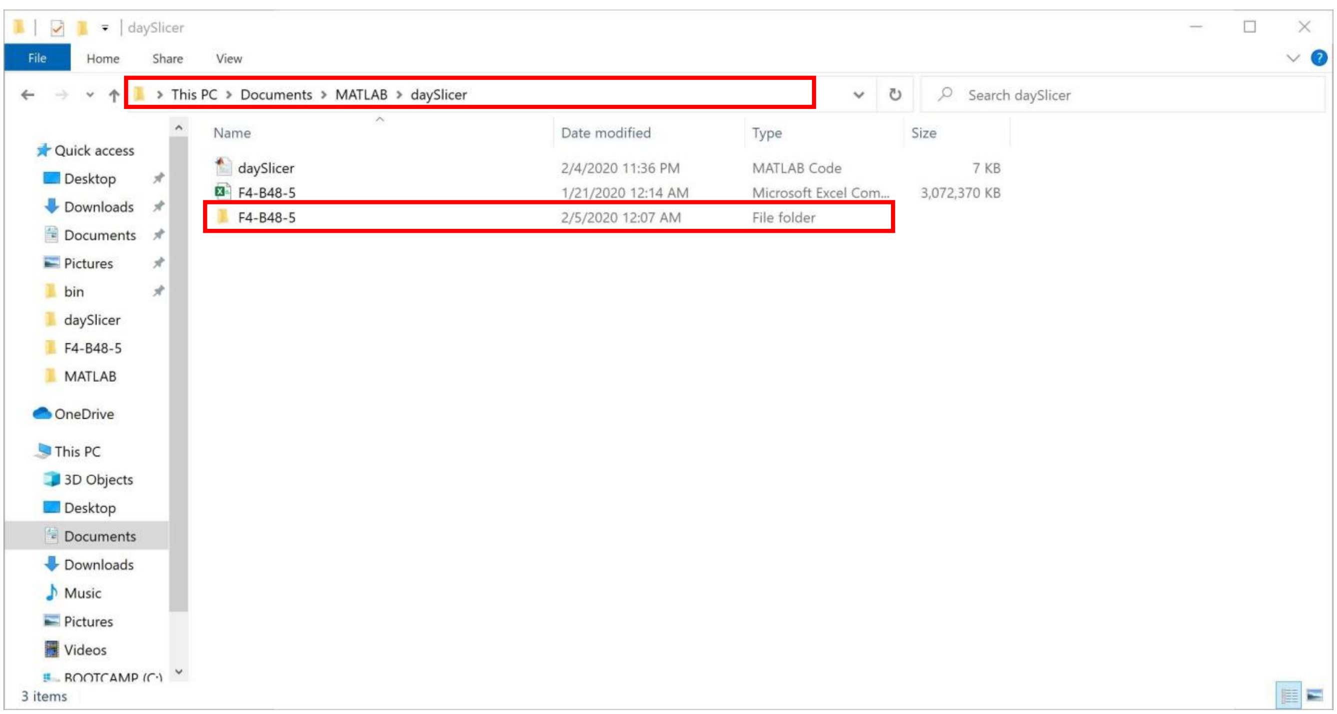
Note: As the sensors we used were made in the UK and used the European date format as yyyy-mm-dd, our functions follow the same trend to prepare and process data.

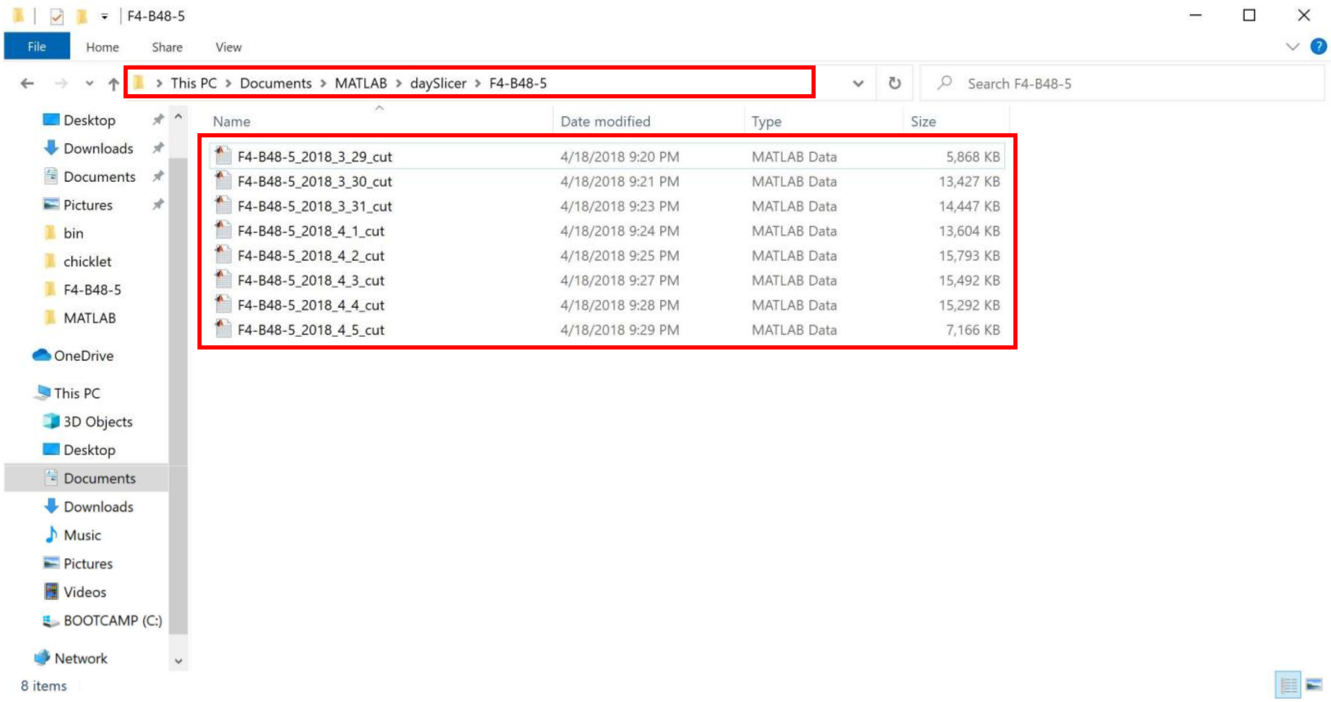


The following would run the daySlicer function on the CSV file starting from 2018-03-29 for 8 consecutive days.



After successful execution of daySlicer you should see a new folder with the same name as CSV file.

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F4-B48-5_2018_4_3_cut 4/18/2018 9:27 PM MATLAB Data 15,492 KB
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F4-B48-5_2018_4_5_cut 4/18/2018 9:29 PM MATLAB Data 7,166 KB

2. *chicklet*:

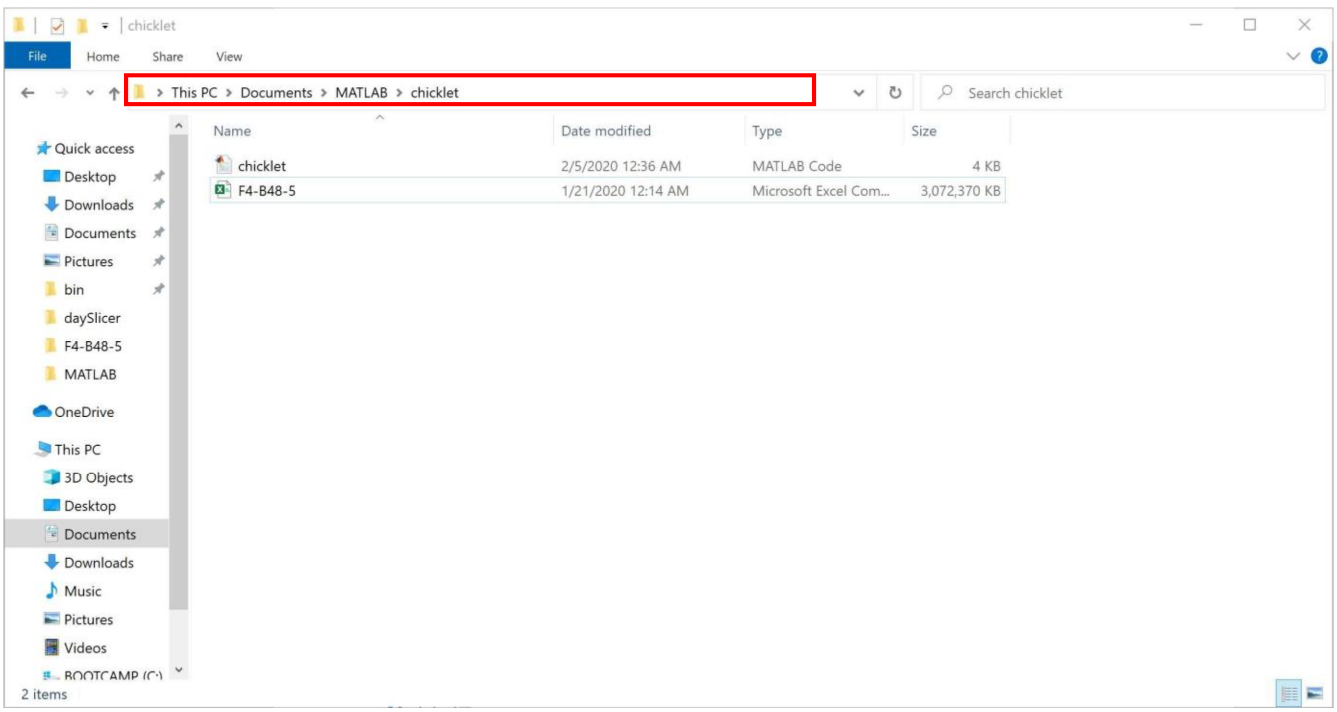
This function allows the user to extract specific parts of the data. Chicklet is similar in functionality to daySlicer function. *daySlicer* slices data as small as individual days (24-hours), whereas, chicklet function can extract much smaller chunks of data (i.e. minutes or hours) with specific starting point in time. The "*chicklet.m*" function is invoked as follows:

chicklet ('2017-07-18 17:50:52', 11, 50)

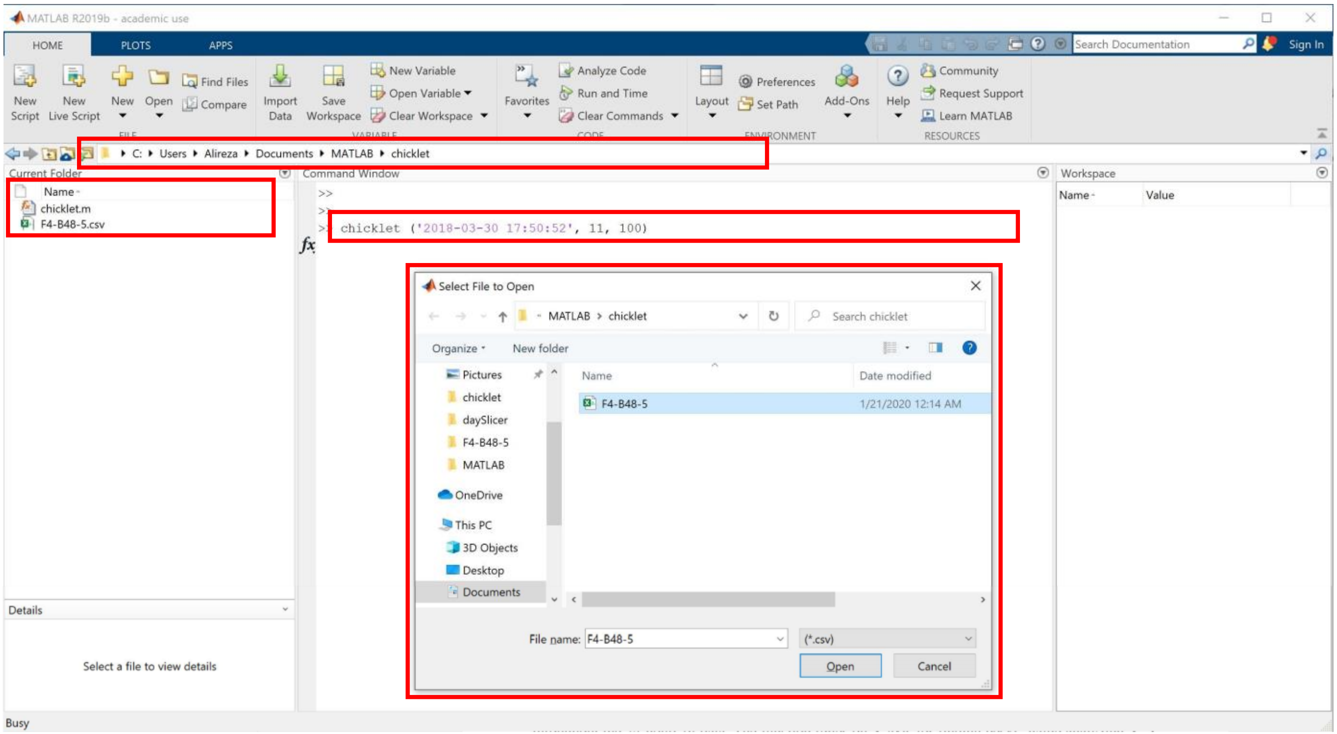
Input: *(1) Starting time-stamp (**Required**) (2) Number of minutes to extract after the starting timestamp (**Required**) (3) The recording frequency of the sensor (**Optional**)(**Default: 100Hz**).*

Output: *The extracted data will be saved as a CSV file in the same directory with the initial timestamp as part of the file name to make it easier to recognize the files.*

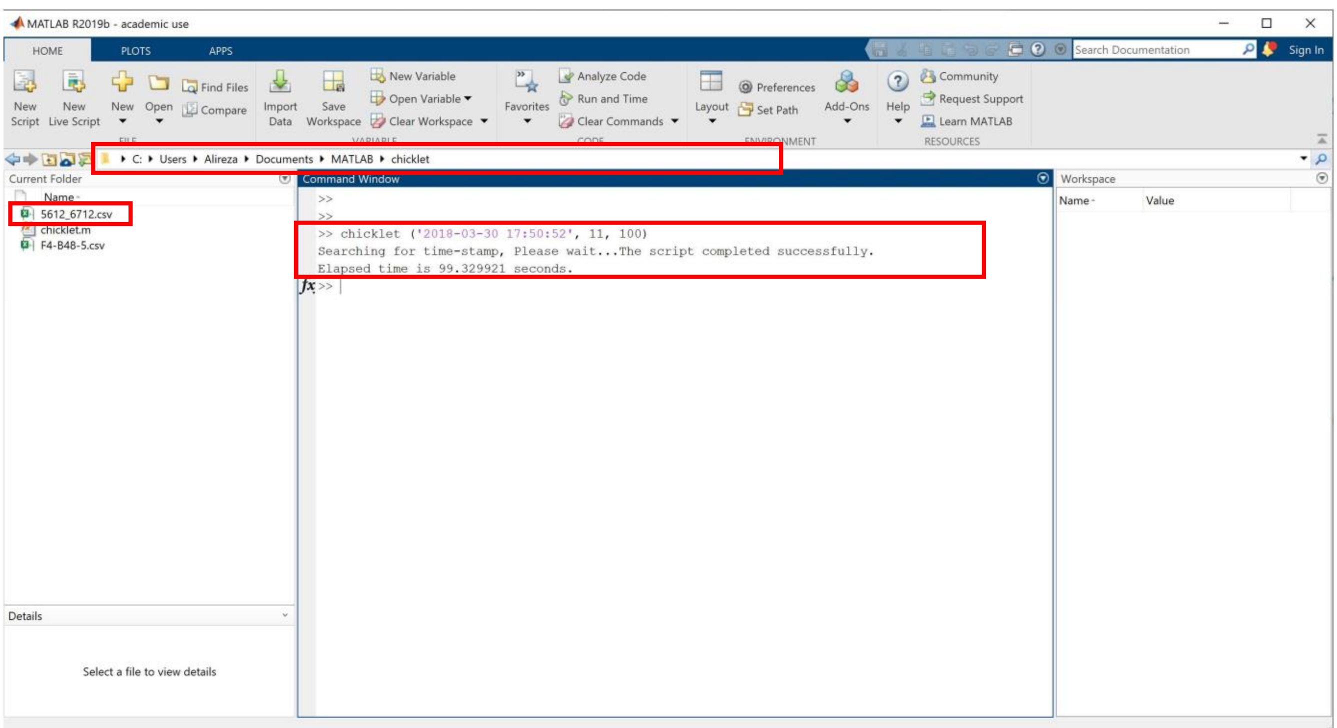
Please put the "*chicklet.m*" function along with the CSV file in the same folder. Change your MATLAB path to the newly created folder and run the *chicklet* function as discussed above.



After you run the chicklet function a prompt window opens up and asks for the input CSV file, as shown below:



Following completion of chicklet function the extracted chunk of data can be found in the same folder as the original CSV file.



3. *behaviorSearch*:

The "***behaviorSearch.m***" function takes in an already prepared chicken dataset, searches for ***pecking***, ***preening*** and ***dustbathing*** behaviors of interest in chickens.

The function returns the exact location of behavior instances throughout the data. The "***behaviorSearch.m***" function is invoked as follows:

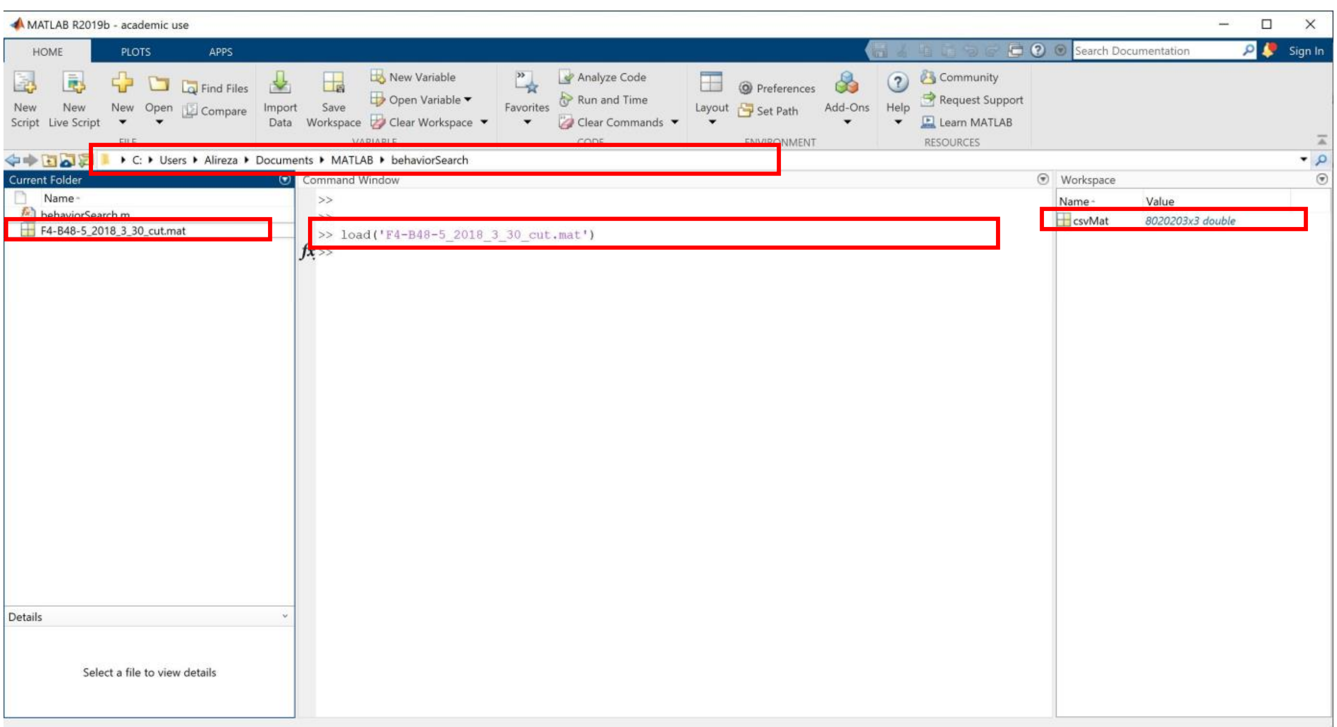
labels = behaviorSearch(chicken_timeseries)

Input: *(1) Three-dimensional (X, Y, and Z axis) time series (**Required**)*

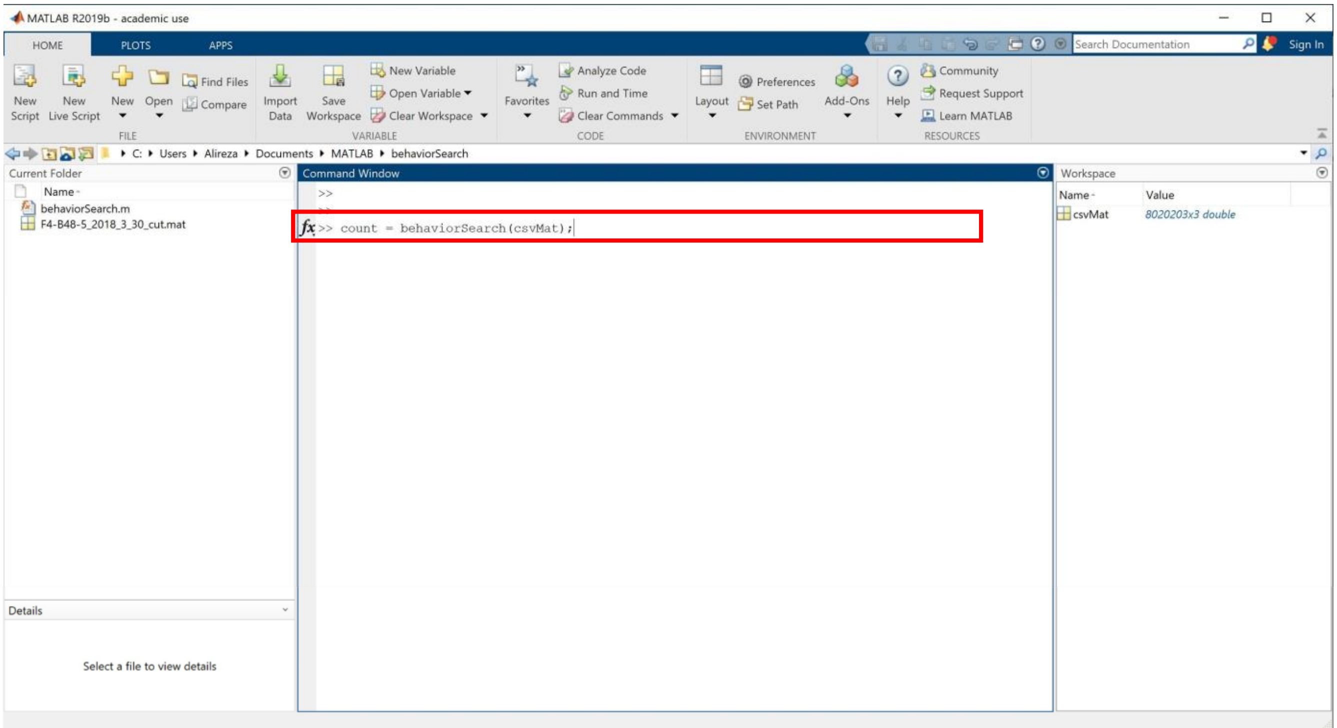
Output: *A vector the same size as the original time series showing the exact region per behavior instance of pecking, preening and dustbathing classes with values 2, 3 and 4 respectively, rest of the regions corresponding to non-behaviors are set to Zero.*

It is better to create a new folder and move the "***behaviorSearch.m***" function and any data files you might want to search for chicken behaviors into the newly created folder. Please make sure you change your MATLAB path to the new folder you just created.

First, you need to load your data files into MATLAB by simply dragging the data file (e.g. from Current Folder on the left side of the picture) onto the MATLAB's command window (the part at the middle of picture). The loaded data will be shown under Workspace section (right side of the picture).



Then, you run the ***behaviorSearch*** function as discussed above and seen in the following. The output label containing exact location of pecking, preening and dustbathing behaviors is assigned to the variable "count" in the following (you are welcome to choose any name for the output variable as you like):



The output data is a vector with values ***0***, ***2***, ***3*** and ***4*** corresponding to ***non-behaviors***, ***pecking***, ***preening*** and ***dustbathing*** instances. In order to get a better intuition of the output visually of the output please print the output vector on top of the original chicken time series data as follows:

```
plot(chicken_timeseries) % Plots the original chicken data in Blue (X), Orange (Y) and red (Z) axis
hold on                  % Allows for plotting on top of already plotted data
plot(labels)             % Plots the vector containing regions of chicken behaviors in purple
```

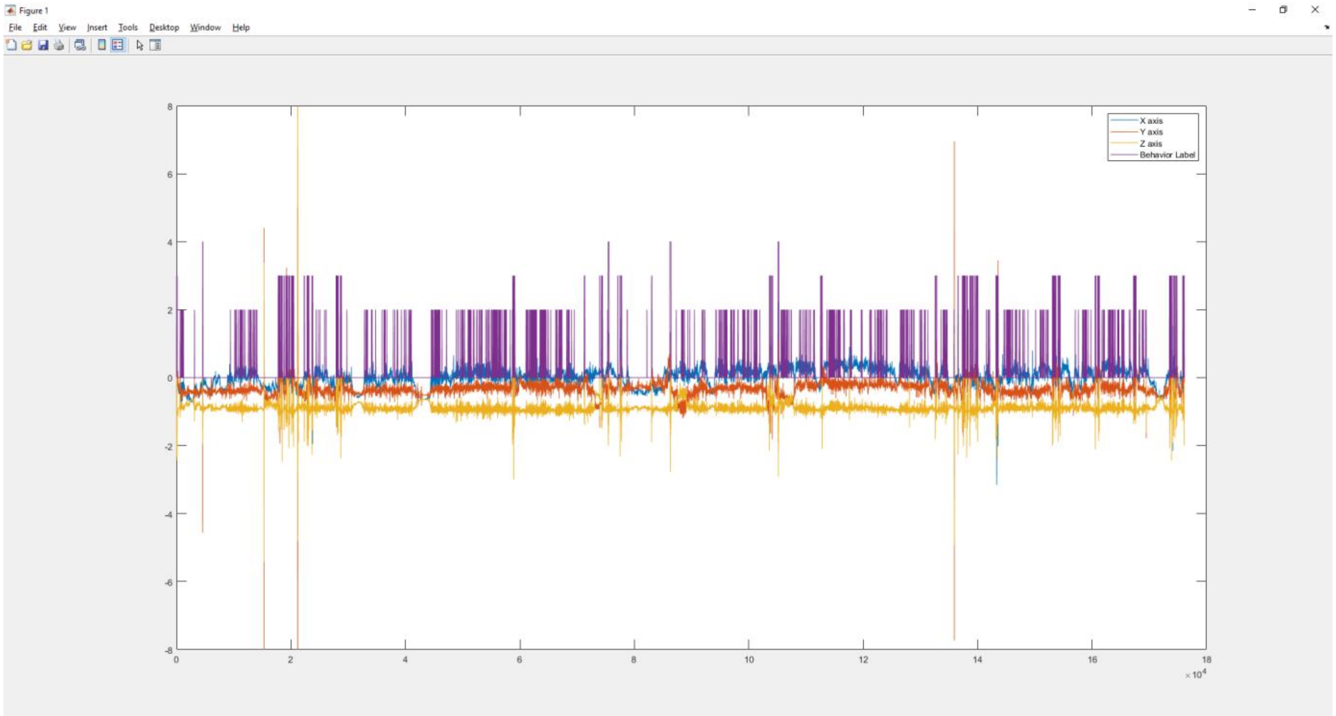


Figure 1
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X axis
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Behavior Label
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12
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18
× 10^4

References

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